



## Technical Reproducibility Validation Report

# STRaitRazor

Version v3.0

STR / SNP panel

|                   |                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------|
| Verification date | 2026-07-06                                                                                              |
| Repository commit | <a href="#">b618e9345ab40f348b504083ae8de2b39abb60fa</a>                                                |
| Attestation level | <b>Runs</b>                                                                                             |
| Permalink         | <a href="https://strhub.app/verified/straitrazor-v3-0">https://strhub.app/verified/straitrazor-v3-0</a> |
| STRhub            | <a href="https://strhub.app">https://strhub.app</a>                                                     |

### SCOPE OF THIS REPORT

Independent automated verification of reproducible execution — confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

#### WHAT WAS VERIFIED

- Source available
- Installation successful
- End-to-end execution
- Output generated

#### NOT VERIFIED

- Genotype accuracy
- Concordance
- Forensic validity
- Regulatory compliance



1. Executive Summary

|                   |                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------|
| PURPOSE           | Verify that the tool installs, runs end-to-end, and produces structurally valid output. |
| RESULT            | 3/5 gates passed — Runs                                                                 |
| ATTESTATION LEVEL | Runs                                                                                    |
| SCOPE             | Installation, execution, and output structure verification.                             |
| NOT EVALUATED     | Genotype accuracy · Concordance · Forensic validity · Regulatory compliance             |

2. Reproducibility Metadata

|                 |                                                                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tool            | STRaitRazor                                                                                                                                           |
| Version         | v3.0                                                                                                                                                  |
| Repository      | <a href="https://github.com/Ahhgust/STRaitRazor">https://github.com/Ahhgust/STRaitRazor</a>                                                           |
| Commit (pinned) | <a href="https://github.com/Ahhgust/STRaitRazor/commit/b618e9345ab40f348b504083ae8de2b39abb60fa">b618e9345ab40f348b504083ae8de2b39abb60fa</a>         |
| Container OS    | ubuntu-22.04                                                                                                                                          |
| Marker panel    | STR / SNP panel                                                                                                                                       |
| Verified on     | 2026-07-06 19:42:24 UTC                                                                                                                               |
| CI run          | <a href="https://github.com/Tfronta/strhub-verified/actions/runs/28818363294">https://github.com/Tfronta/strhub-verified/actions/runs/28818363294</a> |

3. Exact Run Command

The following command was executed verbatim in the CI environment. Replacing the BAM, reference, and BED paths with your own inputs reproduces the verified run.

```
# Environment: ubuntu-22.04 · STRaitRazor v3.0 · commit b618e93
$ /opt/tool/str8rzt \
-c \
/opt/tool/ForenSeqv1.27.config \
/data/in/sample.fastq \
> \
/data/out/sample.allsequences.txt
```

4. Verification Gates

|             |                                                  |      |
|-------------|--------------------------------------------------|------|
| Available   | The pinned public source exists                  | PASS |
| Installs    | The environment builds from source               | PASS |
| Runs        | Executes end-to-end without crashing             | PASS |
| Expected IO | Produces a non-empty file in the declared format | —    |

|                                    |                                                           |   |
|------------------------------------|-----------------------------------------------------------|---|
| <b>Output Structure Validation</b> | Output structure matches expected genotype-bearing format | — |
| Execution log                      | —                                                         |   |

## 5. Out of Scope

This report does not evaluate any of the following:

- Genotype correctness or accuracy
- Concordance against known truth sets
- Sensitivity, specificity, or stutter performance
- Allele calling accuracy or forensic casework suitability
- Regulatory compliance or ISO accreditation
- Multi-laboratory or multi-dataset reproducibility

## 6. Output Content Evidence

|                          |                         |
|--------------------------|-------------------------|
| Output file              | sample.allsequences.txt |
| Format                   | VCF                     |
| Sequence records         | 0                       |
| Distinct STR loci        | 0                       |
| Total reads              | —                       |
| Max depth (single locus) | 0                       |

## 7. Verification Data

This tool does not include its own demo or test data. STRhub ran the verification using the public reference dataset listed below.

|             |                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------|
| Dataset     | NIST mds2-2157 — Illumina STR (ForenSeq slice, donor NTD01)                                         |
| Source      | <a href="https://data.nist.gov/od/id/mds2-2157">https://data.nist.gov/od/id/mds2-2157</a>           |
| License     | research / training / education only (per NIST); not for donor identification or database searching |
| Ref. genome | —                                                                                                   |
| Loci tested | —                                                                                                   |

8. Verification Matrix

|      |                             |                                                             |
|------|-----------------------------|-------------------------------------------------------------|
| PASS | Reference dataset           | NIST mds2-2157 — Illumina STR (ForenSeq slice, donor NTD01) |
|      | README check (5/5)          | Advisory — all items present                                |
|      | Install / environment setup | Present                                                     |
|      | Run command                 | Present                                                     |
|      | Expected input format       | Present                                                     |
|      | Produced output format      | Present                                                     |
|      | Dependencies / versions     | Present                                                     |
|      |                             |                                                             |

9. Limitations

- Single reference dataset per input type
- Single containerized environment (Docker / ubuntu-22.04)
- No truth-set comparison or ground-truth genotypes
- No accuracy or concordance assessment
- No forensic validation of results
- Short-read limitations apply (very long STR alleles may not span reads)

10. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

## 11. Conclusion

### ◆ Runs end-to-end

STRaitRazor v3.0 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit. All 3 verification gates were passed.

### ◆ Produces valid output

The tool generated a structurally valid VCF file with genotype calls across 0 target forensic STR loci, with a maximum read depth of 0 at —.

### ◆ No bundled demo data

STRaitRazor does not include its own test or demo dataset. STRhub Verified supplied the GIAB NA12878 300x dataset (GRCh38) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

### ◆ Reproducibility statement

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.